

Quick Outline of the Fully Explicit Splitting Error Proof

Abstract

This note compresses the full argument into one short chain: pressure perturbation $\Rightarrow$ flux perturbation $\Rightarrow$ transport sensitivity to flux $\Rightarrow$ one-step sequential-versus-fully-implicit bound $\Rightarrow$ global accumulation. It also records the sharper complementary-energy and face-local transport estimators.

1 Setting and notation

Let $\mathcal{C}$ be the set of cells and $\mathcal{F}$ the set of oriented interior faces. Cell i has volume $|C_i|$, and $\mathcal{N}(i)$ is its neighbor set. All pressure vectors lie in the mean-zero subspace so that the discrete pressure operator is invertible.

For a saturation field S , define the TPFA pressure operator by

$$A(S)p = q, \quad (A(S)p)_i := \sum_{j \in \mathcal{N}(i)} T_{ij}(S) (p_i - p_j),$$

where $T_{ij}(S) > 0$ is the face transmissibility. The pressure solution is denoted by $p(S)$. The associated face flux is

$$F_{ij}(S) := T_{ij}(S) (p_i(S) - p_j(S)), \quad F_{ji}(S) = -F_{ij}(S).$$

For a frozen face-flux field F , the explicit upwind transport update is

$$(\mathcal{T}_{\Delta t}^F(S))_i := S_i - \frac{\Delta t}{|C_i|} \sum_{j \in \mathcal{N}(i)} \Phi_{ij}(S; F), \quad \Phi_{ij}(S; F) := F_{ij} f_w(S_{\text{up}}), \quad (1)$$

where $S_{\text{up}} = S_i$ if $F_{ij} \geq 0$ and $S_{\text{up}} = S_j$ otherwise. Assume $f_w : [0, 1] \rightarrow [0, 1]$ is Lipschitz and bounded; write

$$L_f := \sup_{s \in [0, 1]} |f'_w(s)|, \quad \|f_w\|_{\infty} := \sup_{s \in [0, 1]} |f_w(s)|.$$

We use the norms

$$\begin{aligned} \|S\|_{\ell^1(\mathcal{C})} &:= \sum_{i \in \mathcal{C}} |C_i| |S_i|, \\ \|G\|_{\ell^1(\mathcal{F})} &:= \sum_{(i,j) \in \mathcal{F}} |G_{ij}|, \\ \|G\|_{\ell^2(\mathcal{F}; T'^{-1})}^2 &:= \sum_{(i,j) \in \mathcal{F}} \frac{|G_{ij}|^2}{T'_{ij}}, \\ \|p\|_{A(S)}^2 &:= p^T A(S) p = \sum_{(i,j) \in \mathcal{F}} T_{ij}(S) (p_i - p_j)^2, \\ \|r\|_{A(S)^{-1}}^2 &:= r^T A(S)^{-1} r. \end{aligned}$$

2 Convex viewpoint for the pressure problem

Proposition 2.1 (Strong convexity of the pressure functional). *For fixed S , define*

$$J_S(p) := \frac{1}{2} p^T A(S) p - q^T p.$$

Then $p(S)$ is the unique minimizer of J_S on the mean-zero subspace and

$$J_S(p) - J_S(p(S)) = \frac{1}{2} \|p - p(S)\|_{A(S)}^2.$$

If $\lambda_{\min}(A(S)) \geq \underline{\lambda} > 0$, then

$$J_S(p) - J_S(p(S)) \geq \frac{1}{2} \underline{\lambda} \|p - p(S)\|_2^2.$$

Remark 2.2. This convex formulation is not needed for the main splitting estimate, but it explains why the pressure subproblem is well posed and why auxiliary pressure-like solves naturally appear in the estimator.

3 Pressure perturbation and flux perturbation

Fix two saturation states S and S' , and write

$$A := A(S), \quad A' := A(S'), \quad p := p(S), \quad p' := p(S').$$

Also set

$$T_{ij} := T_{ij}(S), \quad T'_{ij} := T_{ij}(S'), \quad \delta T_{ij} := T'_{ij} - T_{ij}.$$

Lemma 3.1 (Pressure perturbation inequality). *The pressure difference satisfies*

$$A'(p' - p) = (A - A')p, \quad \|p' - p\|_{A'} \leq \|(A' - A)p\|_{A'^{-1}}.$$

Proof. Subtract $Ap = q$ and $A'p' = q$ to get $A'(p' - p) = (A - A')p$. Testing with $p' - p$ and applying Cauchy–Schwarz in the A' inner product gives

$$\|p' - p\|_{A'}^2 \leq \|p' - p\|_{A'} \|(A' - A)p\|_{A'^{-1}},$$

which implies the claim. □

Definition 3.2 (Baseline elliptic defect). The defect controlling the flux lag is

$$D(S, S')^2 := 2 \|(A' - A)p\|_{A'^{-1}}^2 + 2 \sum_{(i,j) \in \mathcal{F}} \frac{|\delta T_{ij}|^2}{T'_{ij}} (p_i - p_j)^2. \quad (2)$$

Proposition 3.3 (Flux perturbation bound). *The flux difference satisfies*

$$\|F(S') - F(S)\|_{\ell^2(\mathcal{F}; T'^{-1})} \leq D(S, S').$$

Proof. For each face,

$$F_{ij}(S') - F_{ij}(S) = T'_{ij}((p'_i - p'_j) - (p_i - p_j)) + \delta T_{ij}(p_i - p_j).$$

Squaring, dividing by T'_{ij} , and using $(a+b)^2 \leq 2a^2 + 2b^2$ gives

$$\frac{|F_{ij}(S') - F_{ij}(S)|^2}{T'_{ij}} \leq 2T'_{ij}|(p'_i - p'_j) - (p_i - p_j)|^2 + 2\frac{|\delta T_{ij}|^2}{T'_{ij}}(p_i - p_j)^2.$$

Summing over faces yields

$$\|F(S') - F(S)\|_{\ell^2(\mathcal{F}; T'^{-1})}^2 \leq 2\|p' - p\|_{A'}^2 + 2 \sum_{(i,j) \in \mathcal{F}} \frac{|\delta T_{ij}|^2}{T'_{ij}}(p_i - p_j)^2.$$

Now insert Lemma 3.1. □

Definition 3.4 (Face norm-conversion constant). Define

$$C_{\text{ell}}(S') := \left(\sum_{(i,j) \in \mathcal{F}} T'_{ij} \right)^{1/2}. \quad (3)$$

Then for any face field G ,

$$\|G\|_{\ell^1(\mathcal{F})} \leq C_{\text{ell}}(S') \|G\|_{\ell^2(\mathcal{F}; T'^{-1})}. \quad (4)$$

4 Transport contraction and transport sensitivity to flux

Assumption 4.1 (CFL condition). The explicit transport step satisfies

$$\frac{\Delta t}{|C_i|} \sum_{j \in \mathcal{N}(i)} |F_{ij}| L_f \leq 1 \quad \text{for every } i \in \mathcal{C}. \quad (5)$$

Lemma 4.2 (Crandall–Tartar). *Let $T : \mathbb{R}^{|\mathcal{C}|} \rightarrow \mathbb{R}^{|\mathcal{C}|}$ be order preserving and conservative:*

$$U \leq V \text{ componentwise} \implies T(U) \leq T(V),$$

$$\sum_i |C_i| T(U)_i = \sum_i |C_i| U_i.$$

Then

$$\|T(U) - T(V)\|_{\ell^1(\mathcal{C})} \leq \|U - V\|_{\ell^1(\mathcal{C})}.$$

Remark 4.3. The proof is the usual max/min squeeze: set $M = \max(U, V)$ and $m = \min(U, V)$, use monotonicity to obtain $|T(U)_i - T(V)_i| \leq T(M)_i - T(m)_i$, and then use conservation to collapse the sum.

Proposition 4.4 (IMPES ℓ^1 contraction). *Under the CFL condition (5), the upwind transport map $\mathcal{T}_{\Delta t}^F$ is conservative and order preserving. Hence*

$$\|\mathcal{T}_{\Delta t}^F(U) - \mathcal{T}_{\Delta t}^F(V)\|_{\ell^1(\mathcal{C})} \leq \|U - V\|_{\ell^1(\mathcal{C})}.$$

Proposition 4.5 (Sensitivity of transport to the flux field). *For fixed S^n and two face-flux fields F and $\tilde{F}$,*

$$\left\| \mathcal{T}_{\Delta t}^F(S^n) - \mathcal{T}_{\Delta t}^{\tilde{F}}(S^n) \right\|_{\ell^1(\mathcal{C})} \leq \Delta t L_F \left\| F - \tilde{F} \right\|_{\ell^1(\mathcal{F})}, \quad L_F := 2 \|f_w\|_{\infty}. \quad (6)$$

Proof. For fixed S^n , each numerical flux $\Phi_{ij}(S^n; \cdot)$ is piecewise linear in the scalar argument F_{ij} and Lipschitz with constant at most $\|f_w\|_{\infty}$. Summing the cellwise differences from (1), and noting that each interior face appears in at most two cell balances, gives (6). □

5 One-step splitting estimate

Define the sequential update by

$$S_{\text{seq}}^{n+1} := \mathcal{T}_{\Delta t}^{F(S^n)}(S^n),$$

and let the fully implicit state S_{FI}^{n+1} satisfy the coupled fixed-point relation up to tolerance tol_{FI} :

$$\left\| \mathcal{T}_{\Delta t}^{F(S_{\text{FI}}^{n+1})}(S^n) - S_{\text{FI}}^{n+1} \right\|_{\ell^1(\mathcal{C})} \leq \text{tol}_{\text{FI}}. \quad (7)$$

Theorem 5.1 (One-step sequential versus FI bound). *Let $S' := S_{\text{FI}}^{n+1}$. Then*

$$\|S_{\text{seq}}^{n+1} - S_{\text{FI}}^{n+1}\|_{\ell^1(\mathcal{C})} \leq \Delta t L_F C_{\text{ell}}(S') \mathcal{D}(S^n, S') + \text{tol}_{\text{FI}}. \quad (8)$$

Proof. Introduce the intermediate state

$$\widehat{S} := \mathcal{T}_{\Delta t}^{F(S')}(S^n).$$

By the triangle inequality,

$$\|S_{\text{seq}}^{n+1} - S_{\text{FI}}^{n+1}\|_{\ell^1(\mathcal{C})} \leq \|S_{\text{seq}}^{n+1} - \widehat{S}\|_{\ell^1(\mathcal{C})} + \|\widehat{S} - S_{\text{FI}}^{n+1}\|_{\ell^1(\mathcal{C})}.$$

The second term is at most tol_{FI} by (7). For the first term, apply Proposition 4.5:

$$\|S_{\text{seq}}^{n+1} - \widehat{S}\|_{\ell^1(\mathcal{C})} \leq \Delta t L_F \|F(S^n) - F(S')\|_{\ell^1(\mathcal{F})}.$$

Use the norm conversion (4), then Proposition 3.3, to get

$$\|F(S^n) - F(S')\|_{\ell^1(\mathcal{F})} \leq C_{\text{ell}}(S') \mathcal{D}(S^n, S').$$

Combining the two pieces yields (8). □

Remark 5.2. In the baseline estimate, the direct transmissibility-change term in (2) is part of the defect and should remain. It can only be removed if one replaces the baseline estimate by the sharper complementary-energy route below.

6 Global accumulation and outer iteration

Corollary 6.1 (Global accumulation without Grönwall). *Let*

$$E_n := \|S_{\text{seq}}^n - S_{\text{FI}}^n\|_{\ell^1(\mathcal{C})}, \quad \mathcal{D}_n := \mathcal{D}(S_{\text{seq}}^n, S_{\text{FI}}^{n+1}).$$

Then

$$E_{n+1} \leq E_n + \Delta t L_F C_{\text{ell}}(S_{\text{FI}}^{n+1}) \mathcal{D}_n + \text{tol}_{\text{FI},n}.$$

Hence, by summation in time,

$$E_N \leq E_0 + \sum_{n=0}^{N-1} \Delta t L_F C_{\text{ell}}(S_{\text{FI}}^{n+1}) \mathcal{D}_n + \sum_{n=0}^{N-1} \text{tol}_{\text{FI},n}. \quad (9)$$

No discrete Grönwall factor appears because the transport map is ℓ^1 -contractive.

Proposition 6.2 (Outer-iteration stopping criterion). *Suppose the SFI outer iteration*

$$S^{n+1,(k+1)} := \mathcal{T}_{\Delta t}^{F(S^{n+1,(k)})}(S^n)$$

is contractive with rate $\rho < 1$ in $\ell^1(\mathcal{C})$. Then the distance to its fixed point $S^{n+1,}$ satisfies*

$$\left\| S^{n+1,(k)} - S^{n+1,*} \right\|_{\ell^1(\mathcal{C})} \leq \frac{\rho}{1-\rho} \left\| S^{n+1,(k)} - S^{n+1,(k-1)} \right\|_{\ell^1(\mathcal{C})}.$$

Thus the outer-iterate increment gives a practical stopping rule.

7 Sharper estimators: complementary energy and face-local transport

The baseline estimate loses sharpness in two places: (i) Young's inequality in the flux decomposition and (ii) the global norm conversion from $\ell^2(\mathcal{F}; T'^{-1})$ to $\ell^1(\mathcal{F})$. These losses can be reduced by working directly in flux space.

Definition 7.1 (Complementary energy). For fixed S' , define the resistance functional

$$\mathcal{J}_{S'}^*(G) := \frac{1}{2} \sum_{(i,j) \in \mathcal{F}} \frac{G_{ij}^2}{T'_{ij}}$$

on the affine space of face fields satisfying $\text{div } G = q$. The true flux $F(S')$ minimizes $\mathcal{J}_{S'}^*$ on this space.

Theorem 7.2 (Exact complementary-energy identity (EST4)). *Let $F := F(S)$ and $F' := F(S')$, and assume $\text{div } F = \text{div } F' = q$. Then*

$$\frac{1}{2} \|F' - F\|_{\ell^2(\mathcal{F}; T'^{-1})}^2 = \mathcal{J}_{S'}^*(F) - \mathcal{J}_{S'}^*(F'). \quad (10)$$

Moreover,

$$\mathcal{J}_{S'}^*(F) = \frac{1}{2} \sum_{(i,j) \in \mathcal{F}} \frac{T_{ij}(S)^2}{T_{ij}(S')} (p_i(S) - p_j(S))^2,$$

which is computable from the old pressure and the new transmissibilities.

Proof. Since F' minimizes $\mathcal{J}_{S'}^*$ under the constraint $\text{div } G = q$, the gradient of $\mathcal{J}_{S'}^*$ at F' is a pressure gradient. Its pairing with the divergence-free perturbation $F - F'$ vanishes by discrete summation by parts. Because $\mathcal{J}_{S'}^*$ is quadratic, the resulting Bregman divergence is exactly $\frac{1}{2} \|F' - F\|_{\ell^2(\mathcal{F}; T'^{-1})}^2$. $\square$

Corollary 7.3 (A priori complementary-energy upper bound). *Before solving the new pressure problem one still has*

$$\frac{1}{2} \|F' - F\|_{\ell^2(\mathcal{F}; T'^{-1})}^2 \leq \mathcal{J}_{S'}^*(F),$$

since the minimizer value $\mathcal{J}_{S'}^*(F')$ is nonnegative.

Proposition 7.4 (Face-local transport estimate (EST5)). *Let $\delta F_{ij} := F_{ij}(S^n) - F_{ij}(S')$ and let $S_{\text{up},ij}^n$ denote the upwind saturation at face (i,j) for the transport step from S^n . Then*

$$\left\| \mathcal{T}_{\Delta t}^{F(S^n)}(S^n) - \mathcal{T}_{\Delta t}^{F(S')}(S^n) \right\|_{\ell^1(\mathcal{C})} \leq 2\Delta t \sum_{(i,j) \in \mathcal{F}} |f_w(S_{\text{up},ij}^n)| |\delta F_{ij}|. \quad (11)$$

Theorem 7.5 (Sharper splitting bound). *Under the hypotheses of Theorem 5.1,*

$$\|S_{\text{seq}}^{n+1} - S_{\text{FI}}^{n+1}\|_{\ell^1(C)} \leq 2\Delta t \sum_{(i,j) \in \mathcal{F}} |f_w(S_{\text{up},ij}^n)| |F_{ij}(S^n) - F_{ij}(S_{\text{FI}}^{n+1})| + \text{tol}_{\text{FI}}. \quad (12)$$

The facewise flux error can then be controlled in three ways:

1. *exact a posteriori via the identity (10);*
2. *a priori via the upper bound $\mathcal{J}_{S'}^*(F)$;*
3. *very cheaply by keeping only the transmissibility-change part of the face-flux decomposition.*

8 Proof logic at a glance

The full argument can be remembered as the following six-step chain:

1. Solve the old pressure problem and compute the old flux $F(S^n)$.
2. Compare old and new pressures through the defect equation $A'(p' - p) = (A - A')p$.
3. Convert pressure error into weighted flux error using the face-flux decomposition.
4. Convert weighted flux error into transport error using flux-Lipschitz continuity of the upwind step.
5. Introduce the intermediate state $\hat{S} = \mathcal{T}_{\Delta t}^{F(S_{\text{FI}}^{n+1})}(S^n)$ and split the error into flux lag plus FI residual.
6. Accumulate in time using Crandall–Tartar contraction, which prevents past transport errors from amplifying.